

【Practice of Social Media Analytics】 HW1 – Link Prediction

<Kaggle> <https://www.kaggle.com/t/a41c85f9008348a2a1e49696e26b443b>

說明：

這次作業是要預測node pair (node1, node2)是否存在隱藏關聯(hidden edge)。會提供大家約32000條的edge, 讓大家建立出social network以及訓練資料集。資料集是一個「有向網路」, 因此每組node pair代表一條具有方向性的邊, 例如(29237, 16563, 1) 代表節點29237指向節點16563有邊, (32869, 24548, 0) 代表節點32869指向節點24548沒有邊。

測試資料集中包含 label=0 及 label=1 的node pair, 總共8000筆, 這是你們要預測的關係。可以使用任何的方法預測, 直接使用呼叫的function也行, 但即使是直接使用現成套件, 請還是要了解你所使用的方法, 並且在書面報告裡簡單介紹。**預測結果要上傳到Kaggle評分, 寫完後要繳交程式檔和書面報告到Moodle。**書面報告中請簡述演算法流程, 並說明如何執行你的程式。

作業成績評分方式: 60%是performance(Kaggle分數), 40%是report

train.csv 是拿來training的資料

node1	node2	label
29237	16563	1
32869	24548	0
48837	17831	1
31387	44509	0
15102	10271	1
33467	34453	0

test.csv 是要predict是否正確的node pair (node1, node2)

idx	node1	node2
0	5416	45023
1	6681	40749
2	44162	40953
3	51387	51233
4	6498	24093
5	22823	43971
6	38179	19379

sample_submission.csv 是上傳到Kaggle的格式

idx	ans
0	
1	
2	
3	
4	
5	
6	

idx是單純的index, 從0開始依序排到7999, ans就是預測結果0或1。(0表示你預測此node pair沒有hidden edge, 1表示你預測此node pair有hidden edge)

Description :

This assignment is to **predict whether a node pair (node1, node2) has hidden relation (i.e., hidden edge)**. There are about 32,000 edges provided for you to reconstruct the social network and the training dataset. This is a directed network, so each node pair represents a directed edge. E.g., (32869, 24548, 0) represents node 32869 to node 24548 have no edge, while (29237, 16563, 1) represents an edge from node 29237 to node 16563.

Testing dataset includes 8,000 node pairs whose labels are 0 or 1. These are the relationships you are asked to predict. You can use any prediction method, and you can use any functions/libraries/packages directly. However, even if you use the existing packages, please do understand the method you use and briefly describe it in your report. **The result should be uploaded to the Kaggle platform for evaluation. You also need to upload the program files and the report document to Moodle.** In your report, please briefly describe the algorithm you use, and provide instructions about how to execute your program.

Homework scoring method: 60% is performance (Kaggle score), 40% is report

train.csv is the information used for training

node1	node2	label
29237	16563	1
32869	24548	0
48837	17831	1
31387	44509	0
15102	10271	1
33467	34453	0

test.csv is the node pair (node1, node2) to be predicted

idx	node1	node2
0	5416	45023
1	6681	40749
2	44162	40953
3	51387	51233
4	6498	24093
5	22823	43971
6	38179	19379

sample_submit.csv is the format uploaded to Kaggle

idx	ans
0	
1	
2	
3	
4	
5	
6	

idx is index, starting from 0 to 7999, **ans** is the prediction result 0 or 1. (0 means there is no hidden edge for this node pair; otherwise the ans is 1.)